

Parameters for a Deterministic Welfare Simulation of a Hy-Line Brown Cage-Free Flock

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Scope and modeling conventions

This report separates **breed-standard targets** from **welfare-research findings**. Breed-standard targets come from Hy-Line Brown performance standards and management material. Welfare-research findings come from peer-reviewed aviary, non-cage, and layer-environment studies, which are often not Hy-Line-specific and therefore should be treated as transfer calibrators rather than genetic constants. Hy-Line's 2024 **Alternative Systems** performance table is the closest standard source for a cage-free flock, while the welfare and air-quality coefficients mostly come from commercial aviary and non-cage studies in brown and white hybrids. ¹

For coding, the most robust deterministic structure is a **target layer** driven by Hy-Line standard curves for hen-day production, cumulative mortality, feed intake, and baseline water intake; a **modifier layer** for ammonia, heat stress, lesions, and feather damage; and a **state-update layer** where environmental and welfare states feed back into production, intake, and mortality. Where the literature gives only partial coefficients, a **piecewise-linear or monotone cubic interpolation** through cited knot points is safer than overfitting a closed form. That is especially true for breed-standard production curves and for lesion and feather trajectories. ²

Breed-standard production and intake targets

Hy-Line Brown **Alternative Systems** standards provide weekly ranges for hen-day production, cumulative mortality, feed intake, and water intake from 18 to 100 weeks. For a deterministic model, the cleanest implementation is to take the midpoint of each published weekly range and interpolate between weeks. Selected knot points are below. ³

Age wk	Hen-day % midpoint	Cumulative mortality %	Feed g/bird/day midpoint	Water mL/bird/day midpoint
18	4.4	0.05	80.5	143
21	71.0	0.20	100.0	176
23	92.3	0.34	107.5	189
25	95.2	0.46	115.5	203
30	95.7	0.71	121.0	213

Age wk	Hen-day % midpoint	Cumulative mortality %	Feed g/bird/day midpoint	Water mL/bird/day midpoint
40	94.0	1.24	120.0	211
60	89.0	2.57	120.0	211
72	84.2	3.73	120.0	211
80	79.3	4.93	120.0	211
90	74.4	6.45	120.0	211
100	70.8	8.40	120.0	211

These midpoint values are direct mid-range reductions of the Hy-Line weekly standard tables for alternative systems. Feed and water are nearly flat after peak lay in the standards, with feed centered around **116–124 g/day** and water around **174–248 mL/day** at normal house temperatures. Hy-Line's North America guide makes the same point and explicitly notes that the published water values are for **normal environmental temperatures of 70–81°F**, and that water use can increase substantially above that range. ⁴

Two coding-ready approximations work well. For hen-day production, a fast-rise/slow-decline approximation is:

```
HDEP_target(age) =
  95.3 / (1 + exp(-1.28*(age - 20.15)))      for age <= 28
  96.0 - 0.116*(age - 28) - 0.00347*(age - 28)^2  for age > 28
```

This is **not** a published Hy-Line formula. It is an engineering fit to the Hy-Line standard table and reproduces the onset-of-lay rise and post-peak decline reasonably well. If you want to stay closest to the standard, use direct interpolation of the weekly Hy-Line midpoints instead. ³

For cumulative mortality, one global analytic formula is less reliable than direct interpolation because the Hy-Line curve is shallow early and accelerates late. The safer implementation is simply:

```
Mortality_target(age) = monotone interpolation through weekly Hy-Line cumulative
mortality values
```

That retains the published standard shape, which rises from **0.05% at 18 weeks** to **8.40% at 100 weeks** in the Alternative Systems table. ³

For water, the best structure is to separate normal-temperature breed baseline from heat response:

```
Water_base(age) = interpolate weekly Hy-Line standard values
Water_actual(age, heat) = Water_base(age) * heat_water_multiplier
```

That preserves the fact that Hy-Line's weekly water standards are normal-environment targets, not hot-house values. ⁵

Atmospheric ammonia dynamics

For a deterministic cage-free model, the most defensible structure is a two-source ammonia balance with **belt manure** and **floor litter** as separate emitters:

$$dC/dt = (E_{\text{belt}} + E_{\text{litter}})/V - ACH*(C - C_{\text{out}})$$

where C is in-house ammonia concentration, V is effective mixed air volume, $ACH = Q/V$, and E_{belt} and E_{litter} are source terms from manure on belts and manure-litter on the floor. This differential equation is a modeling inference, but it is consistent with the ventilation-rate and concentration/emission framework used in layer-house ammonia studies. ⁶

The main empirical point is that **aviaries remain ammonia-sensitive because the litter floor is a persistent source even when the colonies have manure belts**. In the 27-month CSES monitoring study, average daily indoor ammonia was **6.7 ppm** in the aviary house, versus **4.0 ppm** in the conventional cage house and **2.8 ppm** in the enriched colony house. In the aviary, **12 winter days** exceeded **25 ppm**, and the published interpretation was **accumulated floor litter coupled with low building ventilation rate**. The same study found that ammonia concentration was inversely related to ambient temperature and ventilation rate; at **ambient temperature below 10°C**, the aviary's ammonia was significantly higher, while the difference narrowed at warmer conditions because higher ventilation diluted ammonia and promoted drying. ⁷

A strong source-side calibration anchor comes from the Wageningen aviary-ammonia thesis. For aviary litter, the degradation process that generates ammoniacal nitrogen was approximately **4% higher per +1°C litter temperature**, **4% higher per +0.1 pH unit**, and **4% higher per +10 g/kg litter water content**. The same thesis reports a house-level ammonia-emission model in a tiered-wire-floor aviary where ammonia emission changed by approximately **+0.76% per hour of manure-removal interval**, **+8.1% per +1°C indoor temperature**, **+0.32% per +1 g/kg litter water content**, and **+103% per +1 m/s air velocity over the litter**, with a mean emission of **2.85 mg/h per hen** under **daily belt removal**. ⁸

That is strong enough for a coding form like:

$$\begin{aligned} E_{\text{total}} = & E_{\text{ref}} \\ & * \exp(0.0076*(RI_h - RI_{\text{ref}})) \\ & * \exp(0.081*(T_{\text{in}} - T_{\text{ref}})) \\ & * \exp(0.0032*(LWC - LWC_{\text{ref}})) \\ & * \exp(1.03*(v_{\text{litter}} - v_{\text{ref}})) \end{aligned}$$

Use this as a **relative modifier around a calibrated baseline**, not as a universal intercept, because the thesis summary provides sensitivities and a reference mean but not a single all-house intercept. ⁹

A second very useful calibration is the effect of **manure accumulation time on manure belts**. Across commercial conventional cage, enriched colony, and aviary houses with manure removed every **3 to 4 days**, mean ammonia emission rates were reported as **0.061, 0.064, 0.085, and 0.115 g/hen/day** at manure accumulation times of **1, 2, 3, and 4 days**, respectively. Relative to day 1, that is about **1.00, 1.05, 1.39, and 1.89×**. ¹⁰

A simple coding reduction is therefore:

```
f_MAT(days) = {1.00, 1.05, 1.39, 1.89} for days = 1,2,3,4
```

If you need a continuous approximation across 1 to 4 days rather than a lookup table, a reasonable engineering fit is:

```
f_MAT(days) ≈ exp(0.20*(days - 1) + 0.03*(days - 1)^2)
```

That continuous form is an approximation to four observed means, not a published biological law. ¹⁰

On the user's “~10× reduction” point, the literature supports **two different clearing effects**, and it is important not to mix them. **System-level management change** from long in-house manure storage to manure-belt management can produce roughly order-of-magnitude lower emissions. A synthesis of layer-house work reported **316 g/AU-day** for high-rise versus **38 g/AU-day** for manure-belt housing, roughly an **8× emission reduction**, and a system analysis summarized manure-belt house emissions as **less than 10% of high-rise counterparts**. That is the right place to use the “10×” idea. ¹¹

Within belt-equipped houses, however, **same-cycle manure clearance** produces a smaller but still meaningful immediate drop because litter or residue remains. In a summer field study of conventional cage, aviary, and natural-mating houses, ammonia concentration after manure removal fell by **64.3%, 28.6%, and 35.7%**, respectively. In other words, an aviary can show an immediate same-cycle drop after manure clearance, but it will usually **not** collapse by 10× if the floor litter remains as a source. ¹²

A practical deterministic clearing rule is therefore:

```
After belt removal:  
E_belt <- r_clear * E_belt
```

with **r_clear** set from observed housing type. For a cage-free aviary, a first-pass concentration-side value near **0.71** is reasonable because the aviary concentration drop after clearance was **28.6%** in the cited summer study. If litter drying is active and floor manure is strongly controlled, the source can be much lower: with **daily belt removal** and forced litter drying, exhaust ammonia was reported **below 5 ppm** and house emission stabilized around **2.0 mg/h per hen** by about **30 weeks of age**. ¹³

For the **ventilation-clearance timescale**, the best practice is to make the timescale depend on the house's air-turnover rate, not to hard-code one universal number:

```
t_63 ≈ 1 / ACH  
t_90 ≈ 2.3 / ACH
```

This is a modeling inference from the well-mixed balance. The evidence base supports that higher ventilation lowers concentration and that post-removal concentration changes are effectively **same-day / within the next ventilation cycle**, but it does **not** support one universal minute-scale time constant for all aviary buildings. ¹⁴

Heat stress response functions

Hy-Line's heat-stress technical update gives an explicit commercial-layer heat stress index:

```
HSI = 0.6 * dry-bulb temperature + 0.4 * wet-bulb temperature
```

and classifies **Alert** at **70–75** and **Danger** at **76–81**. The same technical note states that water:feed ratio is normally **2:1 at 21°C** and rises to **8:1 at 38°C**. It also recommends keeping drinking water **below 25°C** and warns that water **above 30°C** suppresses intake. ¹⁵

For a deterministic model, the cleanest water-intake spike signature is:

```
WF_ratio(T) =  
    2.0                                if T <= 21  
    2.0 + (8.0 - 2.0)*(T - 21)/(38 - 21) if 21 < T < 38  
    8.0                                if T >= 38  
  
Water(T) = Feed(T) * WF_ratio(T)
```

This is a coding interpolation between two Hy-Line anchors. It should be treated as a **heat-stress signature**, not as a replacement for the weekly baseline water table. Hy-Line's North America guide gives a second, more conservative anchor by noting that above the normal **70–81°F** range, water use can rise **up to double** the standard chart values. ¹⁶

The panting and acute-mortality thresholds are unusually well constrained by the 2020 Frontiers experiment. The study found **no panting** at **THI 25.3°C**, panting responses already by **THI 28.5–29°C** in about **40%** of hens, and essentially **100% panting** once **THI exceeded 30°C**, with panting rates above **200 counts/min**. At around **THI 34°C**, panting exceeded **250 counts/min** and mortality was also high. The paper also notes, citing earlier physiology work, that panting was **not observed at 32°C**, but **did occur at 35°C** with slight alkalosis; at **38°C** panting occurred with increased rectal temperature and moderate alkalosis, and at **41°C** with severe alkalosis. ¹⁷

A clean deterministic panting function is therefore:

```
Panting_fraction(THI) =
  0                                if THI < 28.5
  0.6*(THI - 28.5)/(30.0 - 28.5)  if 28.5 <= THI < 30
  1                                if THI >= 30
```

This is an interpolation through the cited thresholds, not a published equation. If you prefer temperature-only logic, use **panting onset near 35°C** and near-universal panting by **about 38°C** as the simpler field proxy. ¹⁸

For mortality, the same Frontiers study gives very useful acute thresholds. If THI rose from **24.2 to 32.1°C within 1 hour** and stayed high for **4.5 hours**, mortality exceeded **95% by 5 hours** and reached **100% at 5.5 hours**. If THI rose more gradually to **31.2°C over 6 hours**, no mortality occurred in the first **3 hours**. If THI rose from **25.2 to 34.3°C within 3 hours** and stayed elevated, mortality first appeared at **4 hours**, was **75% at 5 hours**, and **79% at 8 hours**. The clear conclusion is that the **rate of THI increase matters almost as much as the absolute THI**. ¹⁹

For emergency mortality modeling, that means a threshold-plus-duration hazard is better than a temperature-only threshold. A practical implementation is:

```
if THI < 30:
    h_heat = 0
else:
    h_heat = base_hazard(THI) * exposure_multiplier(duration, dTHI_dt)
```

If you want one explicit approximation, this is reasonable:

```
h_heat = 0                                if THI < 30
h_heat = 0.02 * (THI - 30)^2              if THI >= 30 and exposure < 2 h
h_heat = 0.02 * (THI - 30)^2 * exp(0.6*(t - 2)) if THI >= 30 and exposure >= 2 h
```

This is an engineering approximation built to mimic the cited qualitative mortality pattern, not a published biological hazard equation. The experimental thresholds are the hard anchor. ¹⁹

For productivity decline, the highest-confidence broad thresholds are that the thermoneutral zone for laying hens is about **19–22°C**, performance begins to worsen above about **24–25°C**, and heat stress at **30°C and above** clearly depresses intake and production. A Lohmann technical note states that the ideal environmental temperature is **18–24°C**, that hens exposed to **30°C** reduce feed intake with smaller eggs and eventually lower egg production, and that **above 38°C** birds can only eliminate heat by severe panting; a broader poultry review and related sources align with thermoneutral values around **19–22°C** and begin performance loss above the mid-20s. ²⁰

A practical production modifier is therefore:

```

Prod_heat_multiplier(T) =
  1.00                      if T <= 24
  1 - 0.01*(T - 24)         if 24 < T <= 30
  0.94 - 0.03*(T - 30)     if 30 < T <= 35
  severe-risk regime       if T > 35

```

The threshold placement is evidence-based; the slopes are engineering approximations and should be recalibrated if you have house-specific data. ²⁰

Structural and contact lesions

For **keel-bone fracture**, the strongest coding signal is that fracture incidence rises steeply during early-to-mid lay, especially around peak egg production. In a longitudinal study of white and brown hens followed to **65 weeks**, about **62%** had broken keels at depopulation; newly diagnosed fractures were temporally linked to the period of highest laying rate, and birds that began lay earlier were more likely to end with fractures. All birds with bumblefoot on both feet had a fracture at depopulation. ²¹

A larger Danish risk-factor study found that delaying onset of lay reduced fracture risk by about **12% per week**, increasing egg weight at onset raised fracture prevalence by **3% per +1 g**, and each **+100 g** body weight reduced fracture prevalence by about **3–4%**. The same study reports that the majority of fractures develop during **weeks 25–50**, with a peak around **35 weeks**, and that non-caged systems had almost **twice** the estimated number of fractures per hen compared with caged systems after accounting for other factors. ²²

In modified aviaries, mean total-fracture prevalence rose from **60.0% at 29 weeks** to **76.0% at 39 weeks** and **86.5% at 49 weeks**. Ramps reduced fracture prevalence at all ages, and fracture prevalence was already much higher by **29 weeks** in pens without ramps. The same paper interprets brown hens as more fracture-prone than white hens in that aviary, plausibly because poorer flight ability and heavier body weight increase impact forces. ²³

For coding, a piecewise accumulation law is more transparent than a single global logistic:

```

KBF_prevalence(age) =
  ~0 before 22–24 wk
  rise at ~1.0 to 1.6 percentage points/week from 25 to 50 wk
  slower rise / plateau thereafter

```

Then modify the weekly increment by onset-of-lay timing, early egg size, body weight, and aviary design:

```

IncKBF *= 0.88^(weeks_delayed_onset)
IncKBF *= 1.03^(egg_weight_onset_g - ref_g)

```

```
InckBF *= 0.97^((body_weight_g - ref_g)/100)
InckBF *= ramp_factor
```

The coefficient multipliers come directly from the Danish risk-factor study; the piecewise age scaffold is an engineering wrapper around the prevalence trajectories. ²⁴

For **footpad lesions**, the non-cage literature shows both chronic accumulation and strong management dependence. In an Austrian field survey of **297** non-cage flocks aged **18–83 weeks**, ulcerative footpad dermatitis had a flock median of **40%** of hens affected, with mean **38.75%**, SD **23.29%**, and range **0–95%**. Risk increased with **age, higher hen weight, larger flock size, higher stocking density, and higher perches**. ²⁵

A 2024 retrospective German laying-hen study reported that first signs were low at the first visits, began increasing around **peak lay at about 28 weeks**, and by end of lay **34.4%** of hens had at least slight alterations. ²⁶

The modified-aviary experiment is especially useful for dynamics. Across **29, 39, and 49 weeks**, overall dermatitis prevalence (Y/N) was **36.5%, 35.4%, and 38.5%**, but the state composition shifted. In the worst treatment, **mild dermatitis** rose from **30.9% to 53.5% to 74.7%**, while **severe dermatitis** fell from **71.7% to 65.2% to 58.0%**. The authors interpret this as severe lesions healing or transitioning into milder chronic lesions over time. Ramps reduced both dermatitis and bumblefoot, and hybrid differences were also clear. ²⁷

That supports a two-compartment lesion model:

```
dMildFPD/dt = alpha_exposure - beta_progress*MildFPD + gamma_heal*SevereFPD
dSevereFPD/dt = beta_progress*MildFPD - gamma_heal*SevereFPD
```

with **alpha_exposure** increasing with wet litter, density, perch pressure, and age. If you need a single summary score instead, model “any dermatitis” as a slow age-linked rise beginning near **28 weeks**, and let a severity index peak earlier under poor hygiene. ²⁸

Feather cover and feather-pecking progression

Feather damage is the welfare outcome that most clearly shows a **mid-lay to late-lay acceleration** in cage-free systems. In the German commercial-aviary study of non-beak-trimmed flocks, visits were at **30–33 weeks, 44–48 weeks, and 62–68 weeks**. Severe plumage damage increased from **3.2% to 32.9% to 57.8%** in one laying period and from **8.13% to 32.5% to 50.4%** in another. Poor litter quality, no free range, insufficient enrichment, no daylight, higher stocking density, shorter perch length, and plumage damage already present during rearing all increased the risk. ²⁹

The same study found that rearing flocks with high stocking density and poor litter quality had more plumage damage already during rearing. During rearing visits, flock plumage damage ranged from **0 to more than 90%**, and the multifactorial model identified **stocking density** and **litter quality** as significant

risk factors by the later rearing visit. The paper also cites earlier work showing that if plumage damage was present during rearing, hens had a **90% probability** of beginning severe feather pecking later. ³⁰

The Midwestern aviary final report shows the same late-lay deterioration. On the figure labeled “Feather scores of laying hens in the aviary houses,” birds with **bare spots larger than 5 cm** were **0%** at peak, **3%** mid-lay, and **52%** late-lay; birds with moderate wear were **0%, 27%, and 32%**, respectively. The companion text says all hens had no to slight wear early in the cycle, but **just over 50%** had at least one bare spot >5 cm late in lay. ³¹

A deterministic progression law that matches the field evidence is:

```
SevereFeatherDamage(age) =  
  near 0 before 30–33 wk  
  +~1.6 to 2.0 percentage points/week from ~32 to ~46 wk  
  +~1.0 to 1.3 percentage points/week from ~46 to ~65 wk
```

Those weekly slopes come directly from the cited commercial-aviary trajectories: roughly **+29 to +30 percentage points** over about **14–15 weeks** from LV1 to LV2, then another **+18 to +25 points** over **17–20 weeks** from LV2 to LV3. ²⁹

A simple implementation is:

```
dFeatherDamage/dt = r0(age)  
                    * f_rearing  
                    * f_litter  
                    * f_free_range  
                    * f_enrichment  
                    * f_density
```

with `f_rearing > 1` if plumage damage existed in rearing, `f_litter > 1` with poor litter quality, `f_free_range < 1` when range is available and used, `f_enrichment < 1` with adequate enrichment, and `f_density > 1` at high stocking density. The cited studies support the direction of each modifier even when exact cross-study coefficients differ. ³²

Coding-ready parameter set and uncertainty

A compact parameter set that stays close to the highest-confidence evidence is:

Domain	Suggested parameterization	Evidence level
Breed targets	Weekly Hy-Line midpoint interpolation for HDEP, cumulative mortality, feed, and water	High

Domain	Suggested parameterization	Evidence level
Water under heat	$\text{Water} = \text{Feed} * \text{WF_ratio}(T)$ with 2:1 at 21°C to 8:1 at 38°C	High
Heat index	$\text{HSI} = 0.6 * T_{db} + 0.4 * T_{wb}$; alert 70–75, danger 76–81	High
Panting onset	none at THI 25.3; variable at ~28.5–29; near-universal above 30	High
Acute mortality	hazard only in acute high-THI regime; very high after several hours once THI is ~32–34 and rising fast	High
Ammonia source	two-source model: belt + litter	High
Belt manure age effect	relative multipliers 1.00, 1.05, 1.39, 1.89 for MAT 1–4 d	High
Aviary NH ₃ anchors	mean 6.7 ppm overall; winter exceedances >25 ppm when litter accumulated and ventilation was low	High
Emission sensitivities	+0.76%/h removal interval; +8.1%/°C; +0.32% per g/kg litter water; +103% per m/s air speed	Moderate
Litter TAN generation	+4%/°C; +4% per 0.1 pH; +4% per 10 g/kg water	Moderate
KBF accumulation	age-linked rise centered in weeks 25–50; modify by onset-of-lay, early egg weight, body weight, ramps	High
FPD accumulation	onset around peak lay; track mild and severe states separately if possible	Moderate
Feather damage	near-zero at 30–33 wk, then strong rise through 44–48 and 62–68 wk	High

The biggest uncertainty is **transportability to Hy-Line Brown specifically** for the welfare variables. The production and intake standards are Hy-Line-specific. The lesion and feather-pecking coefficients are mostly from brown and white commercial hybrids in aviary systems, not from dedicated Hy-Line Brown challenge trials. The safest use is therefore to treat these welfare coefficients as **informative priors for calibration**, not immutable breed constants. ³³

Open questions and limitations

The literature is much stronger on **thresholds and relative effects** than on one universal intercept for every house design. That is especially true for ammonia source terms, because aviary geometry, manure-drying hardware, floor-litter management, and sampling location all matter. The best practice is to calibrate the baseline source strengths to one farm or one house, then apply the cited multipliers for manure age, temperature, litter moisture, ventilation, and removal events. ³⁴

The keel-bone and footpad lesion literature is also heterogeneous in scoring method and genotype. Some studies use post-mortem fracture counts, others palpation; some use ISA Brown or Dekalb White, others

mixed commercial hybrids. Those differences are large enough that the safest deterministic implementation is a **state-transition structure calibrated to your own chosen baseline prevalence**, using the cited age and risk modifiers to shape progression. 35

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